

2.1 Sensitivity (the derivative)

Study Notes · Third Eye Labs
Season 2, episode 1

THE ONE QUESTION

"If I nudge this number, does the score get better or worse — and by how much?"

Asked once per weight, per step, for the entire history of machine learning.

The setup

one weight w
64 real data points
 $\text{score}(w) = \text{mean}((w \cdot x - y)^2)$
start at $w = 2.0 \rightarrow \text{score } 0.5062$

BEFORE YOU START — if you write software but not machine learning

Nothing here needs calculus you remember. If you have ever tuned a timeout by changing it, re-running the load test, and looking at the difference — you have already taken a derivative. It has a name, that is all.

the ML word	what it actually is, in your terms
weight	A number the program tunes itself instead of you setting it.
score / loss	One number for how wrong we are right now. Here: take every prediction, subtract the right answer, square it (so too-high and too-low both count as wrong), average.
derivative	The answer to "if I change this by a tiny amount, how much does the score move, per unit of change?" A rate, like ms of latency per extra thread.
squared error	Why square it? So misses in both directions count, and so big misses count much more than small ones.

The whole episode is one question asked over and over — the same question you ask a profiler.

1. Nudge it and divide

$$h = 0.1$$

$$\text{score}(2.1) - \text{score}(2.0) = -0.02254$$

$$\div 0.1 = -0.2254$$

that ratio is "rise over run" — the thing you already did at school

h	ratio
0.1	-0.22540
0.01	-0.24754
0.001	-0.24975
0.0001	-0.24998
0.00001	-0.25000

The digits stop changing, one at a time. It settles on -0.25 — and that settling number is the derivative. This is a limit actually happening, not a definition.

2. Check it

3. What the sign means

- ratio **negative** $\rightarrow$ score falls as w rises $\rightarrow$ increase w

method	value
shrinking nudge	-0.25000
autograd (PyTorch)	-0.2500000

Agreement to 6 decimals. The number is real, not an artefact of the arithmetic.

- ratio **positive** → score rises as w rises → decrease w
- The sign is the instruction. Walking downhill from $w = 2.0$ lands at $w = 2.9993$ — and the data was really made with $w = 3.0$.

4. The symbol, last

$$dL / dw$$

⚠ Not a fraction. It is the name of the number you just watched settle. If the notation means nothing to you, ignore it — you already have the idea.

5. Advanced — where this lives

- Partial derivative: the same question with the other knobs held still. Stack them and you get the gradient → 2.2
- Autodiff: frameworks never use shrinking nudges — too slow, too noisy. They apply the chain rule symbolically → 2.4
- Finite differences still matter as a *gradient check*: if your hand-written backward pass disagrees with a nudge, the backward pass is wrong.
- In a transformer: this exact question, asked of 124 million weights, on every single training step.

NEXT: with one weight the answer is a *sign*. With two it has to become a *direction* → 2.2, the gradient.